The Digestive System

oral cavity and salivary glands

SGL1

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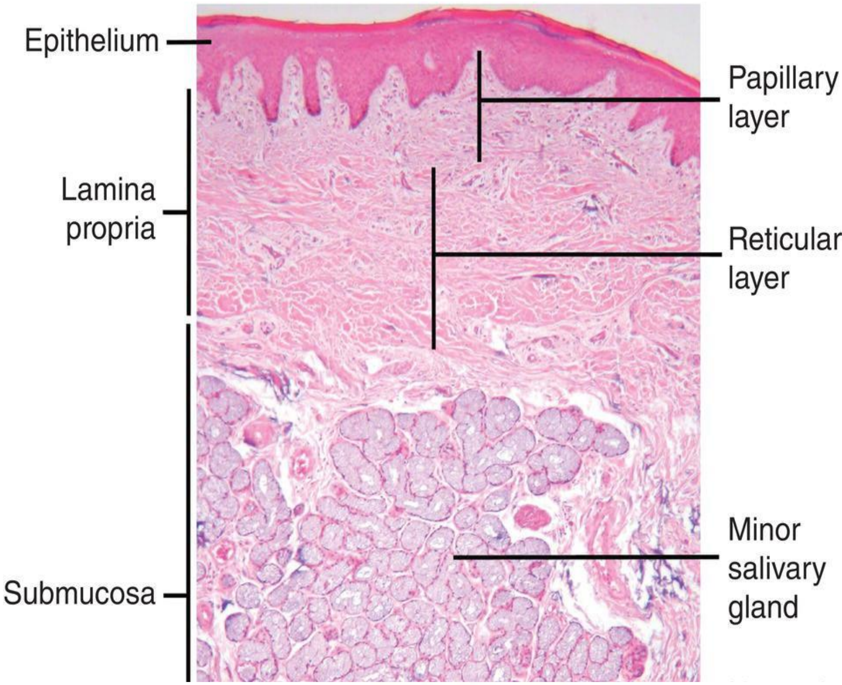
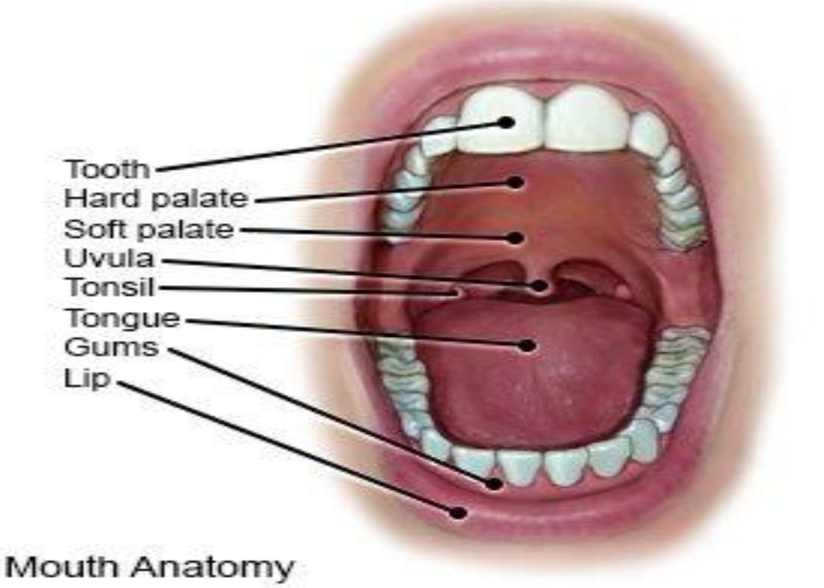
Oral Cavity (mouth): Initial site of mechanical digestion and chemical digestion by(saliva).

Is made up partly of bone (jaws, hard palate) & partly of muscle & connective tissue (lips, cheeks, soft palate & floor of mouth).

- lined by mucous membrane is stratified squamous epithelium which rests on connective tissue.
- In the cheeks, this connective tissue contains many elastic fibers & much fat.
- The roof of the mouth consists of the hard & soft palates, both covered by same type of epithelium.
- In the hard palate the mucous membrane rests on bony tissue. The soft palate has a core of skeletal muscle & numerous mucous glands in its submucosa.





Lips : Is skeletal muscle.•Each lip have an external surface & an internal surface :

The inner surface of the lips lined by thick non keratinized st.sq.ep. & small clumps of salivary gland (labial gland), sebaceous gland.

Outer surface of the lips lined by thin skin with hair follicles, sebaceous glands, and sweat glands.

Between the inner & outer surface of the lips there is a transitional zone known as vermillion border the epithelium is lightly keratinized s.s.ep., contain prominent blood vessels which gives it's red appearance with numerous sensory nerve endings , has no hair ,no sweat & sebaceous glands

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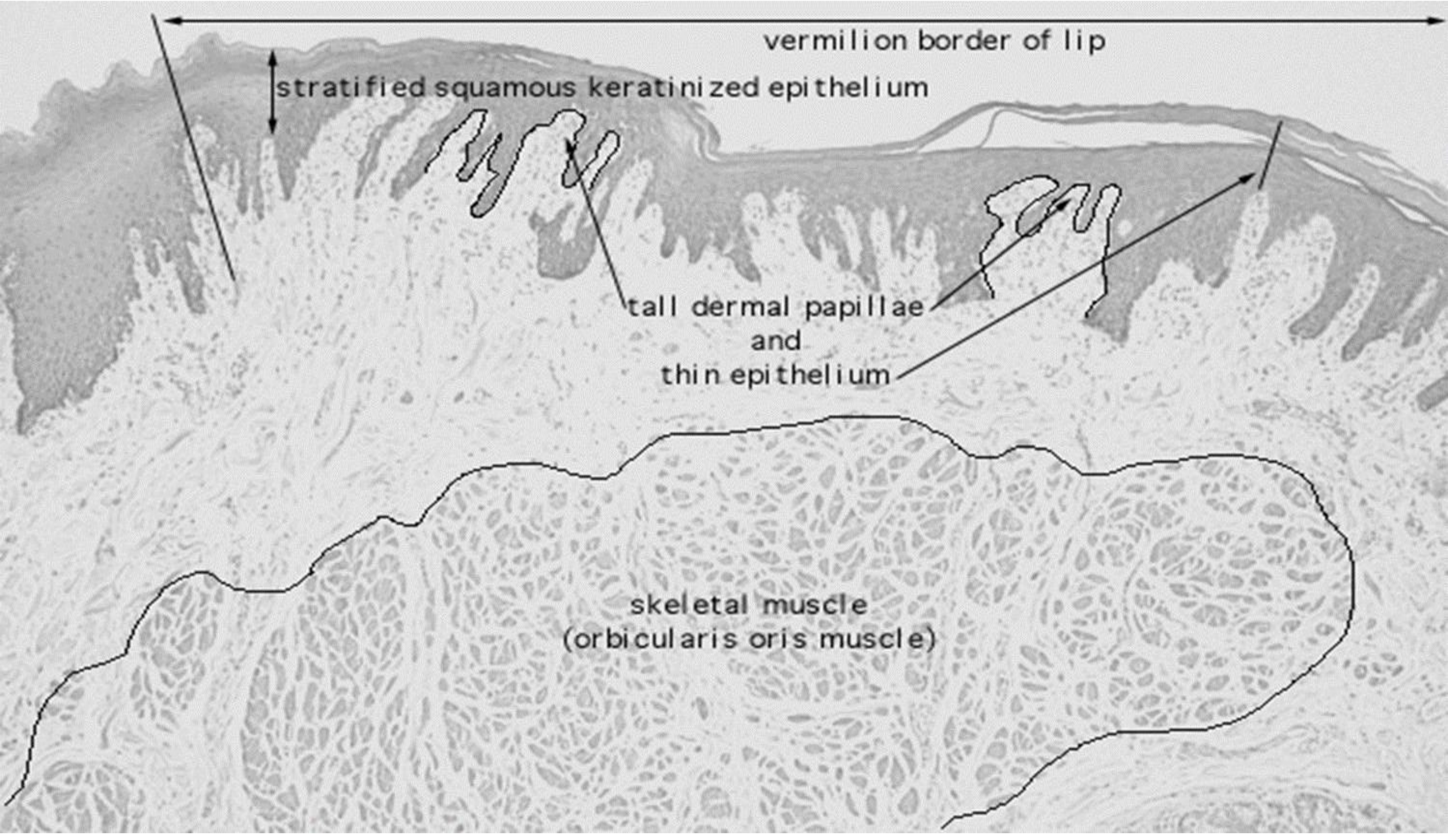
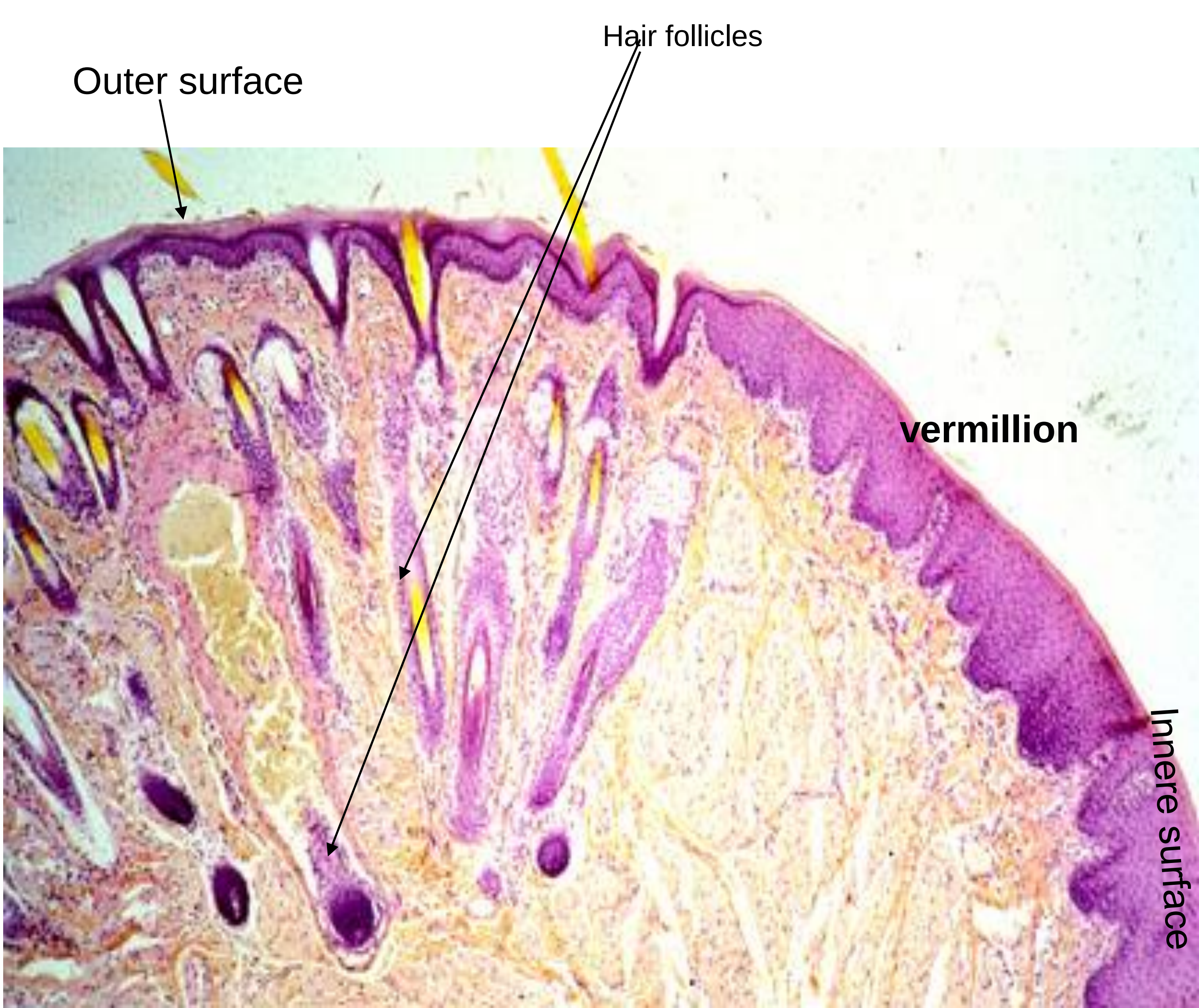
In the deeper parts of lips bundles of striated muscle fibers (orbicularis oris muscle) are arranged mainly in a concentric manner around the oral orifice





Hair follicles
Outer surface
vermillion
Innere surface

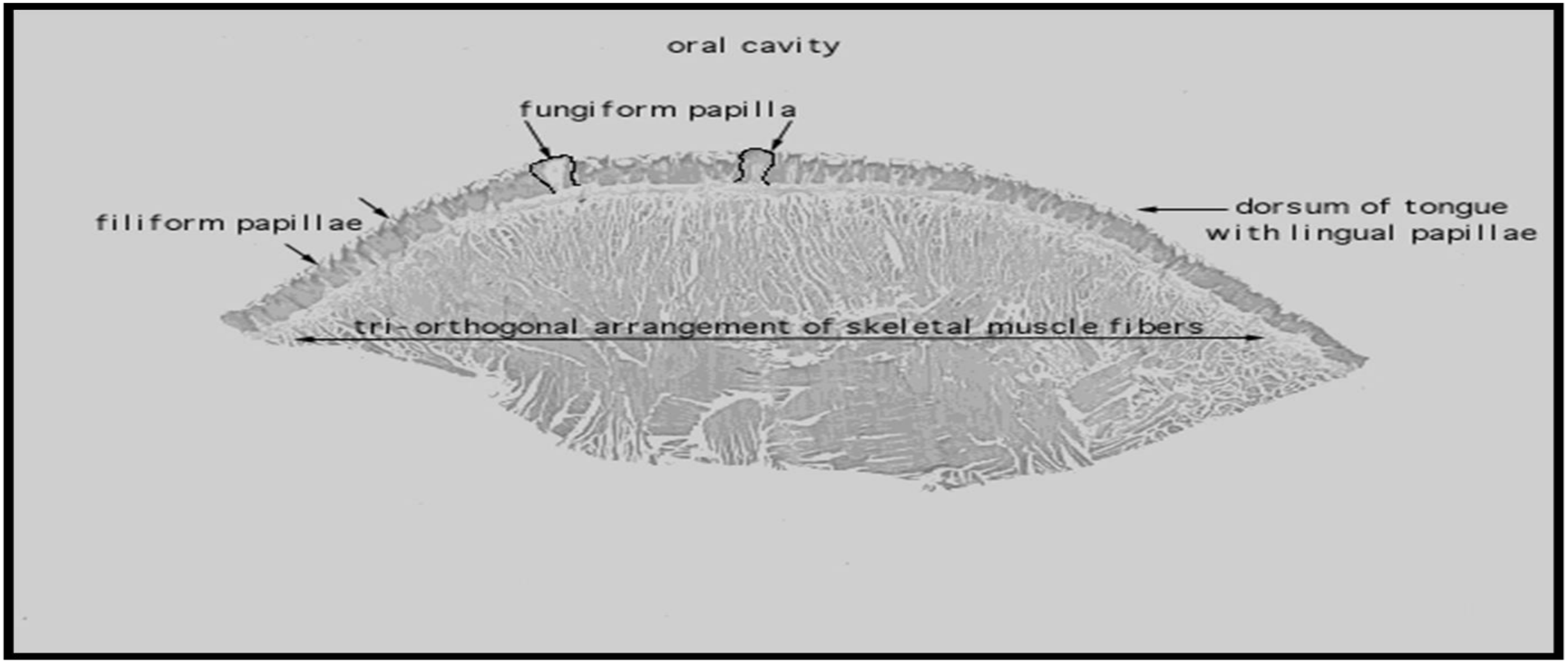
Tongue

. It's a mass of skeletal muscle occupies the floor of the mouth and fills the oral cavity.

- covered with lightly keratinized stratified squamous epithelium.
- The mucosa is bound tightly to the muscle by the lamina propria , the muscle is arranged in bundle of many size these are separated by connective tissue cross in 3 planes arranged in bundle this gives the tongue the flexibility for speech, positioning food , chewing & swallowing

The mucosa differ on the dorsal & ventral surface

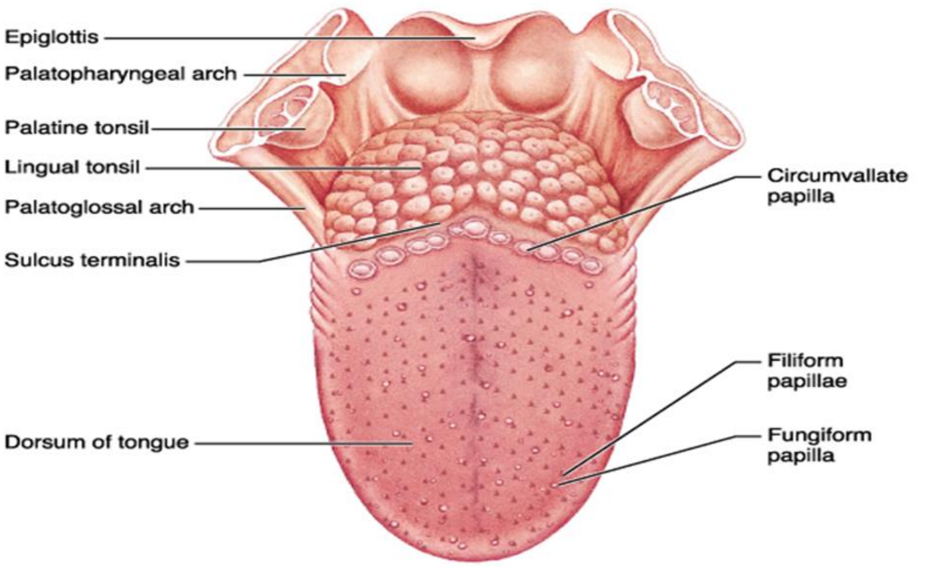
- ventral surface = the floor of the oral cavity covered by non-keratinized s.s.ep.
- dorsal surface is partly keratinized numerous small projections (papillae) cover the superior (dorsal) surface. Each papillae consists of a lining epithelium & a core of connective tissue.
- The anterior two thirds of the dorsal surface is separated from the posterior one third by a V-shaped groove
- Behind this there is lingual tonsils.

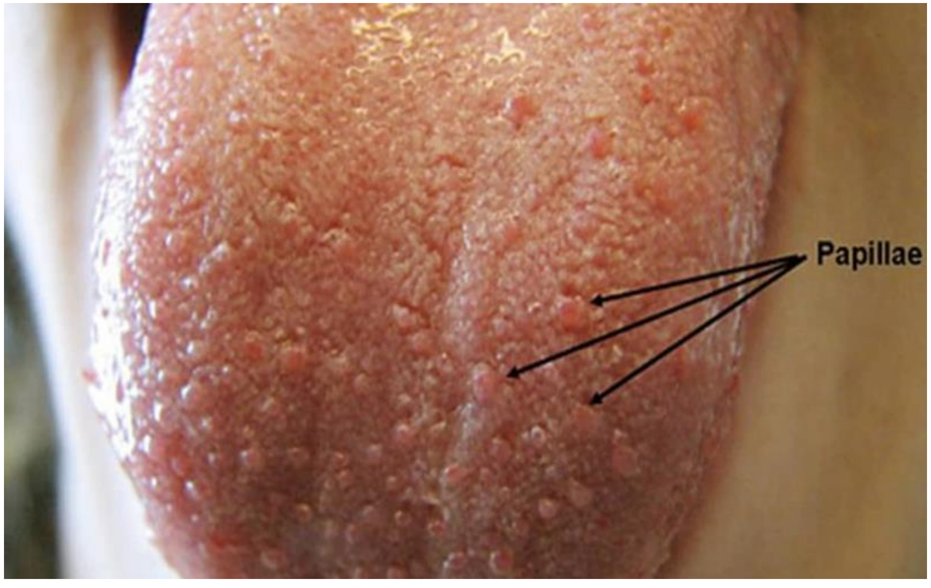


dorsal surface bears four types of papillae

1. Filiform –
2. Fungiform –
3. Circumvallate – V-shaped row in back of tongue
4. Follate

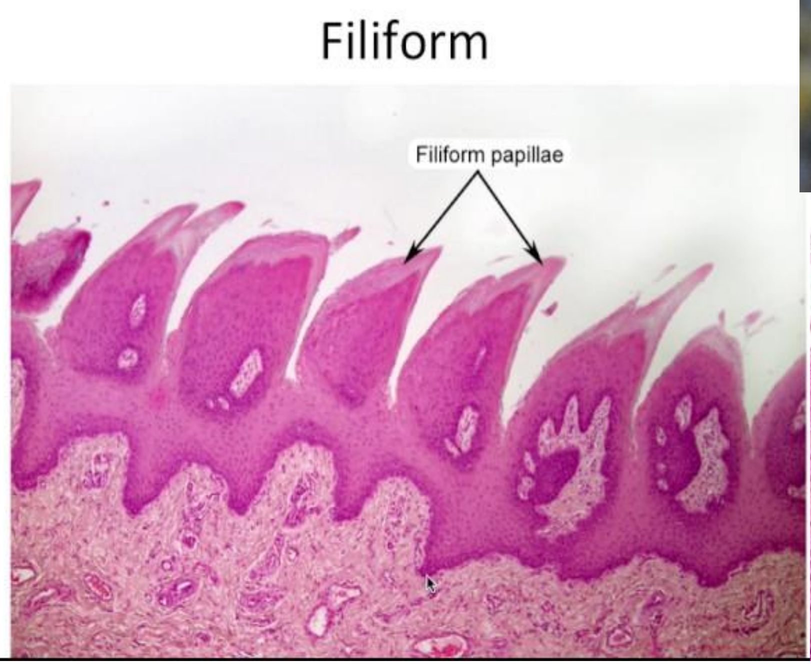
- Sulcus terminalis – groove that separates the tongue into two areas:
  - Anterior 2/3 residing in the oral cavity
  - Posterior third residing in the oropharynx





Filiform papillae

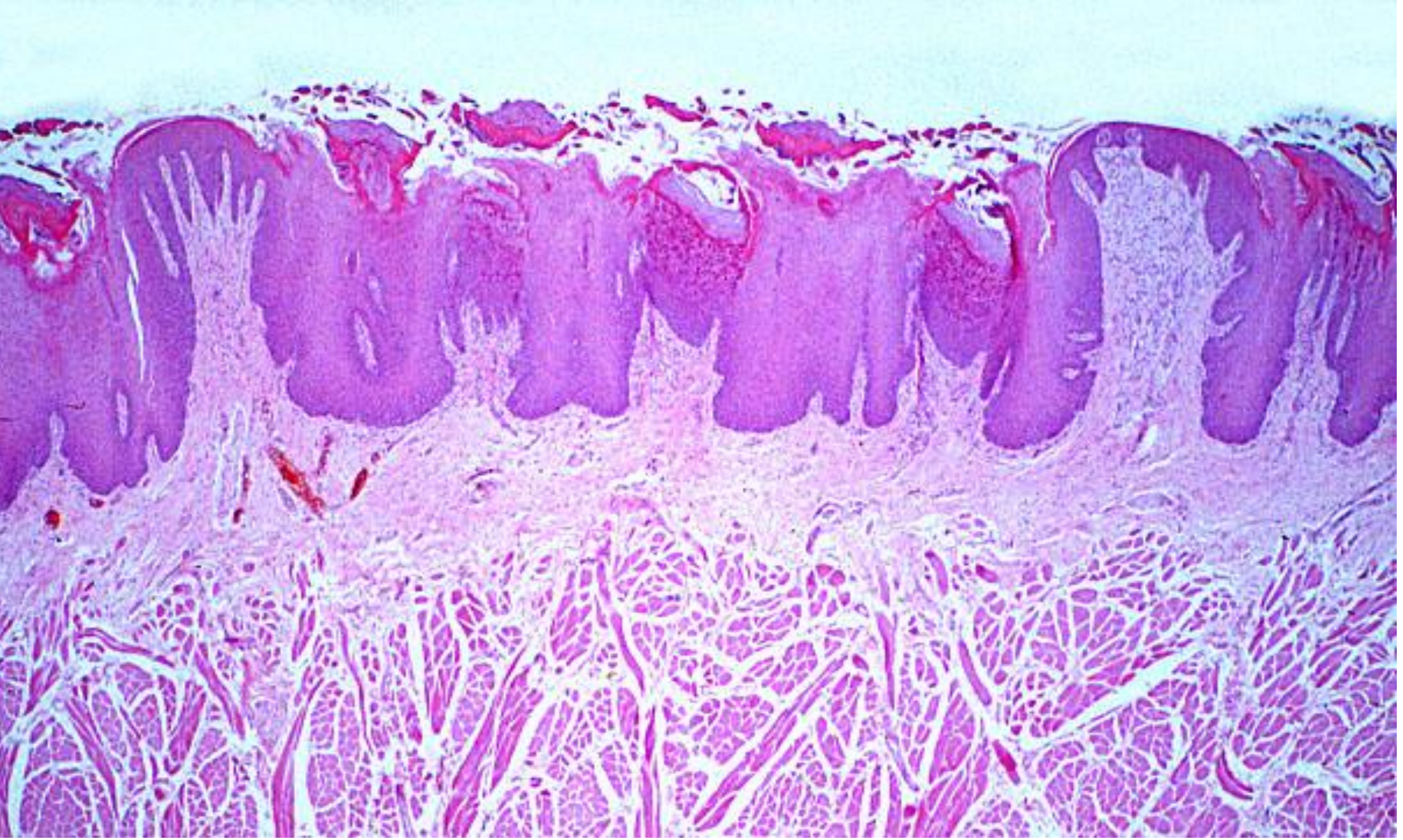
Are **the smallest most numerous elongated conical shape , they present over the entire surface , the epithelium is partly keratinized give the tongue roughness. and provide friction aid in the processing of foods. Not contain taste buds.**



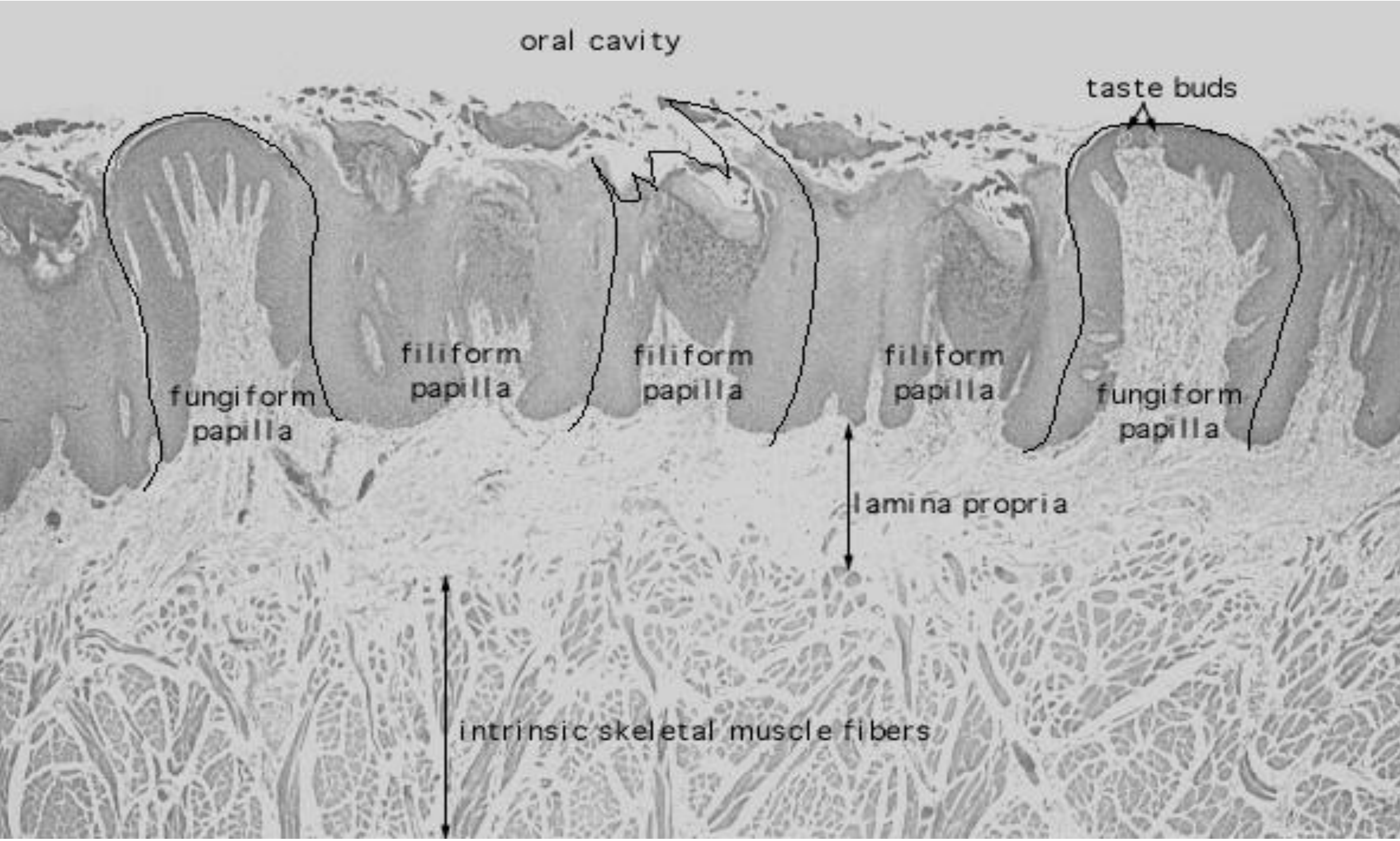
Fungiform papillae

Mashroom shape ,occur singly and are located evenly between the filiform Their connective tissue core is richly vascularised. The epithelium is slightly thinner than on the remaining surface of the tongue. Contain scattered taste buds . Detecting sweet and salty taste.

Tongue



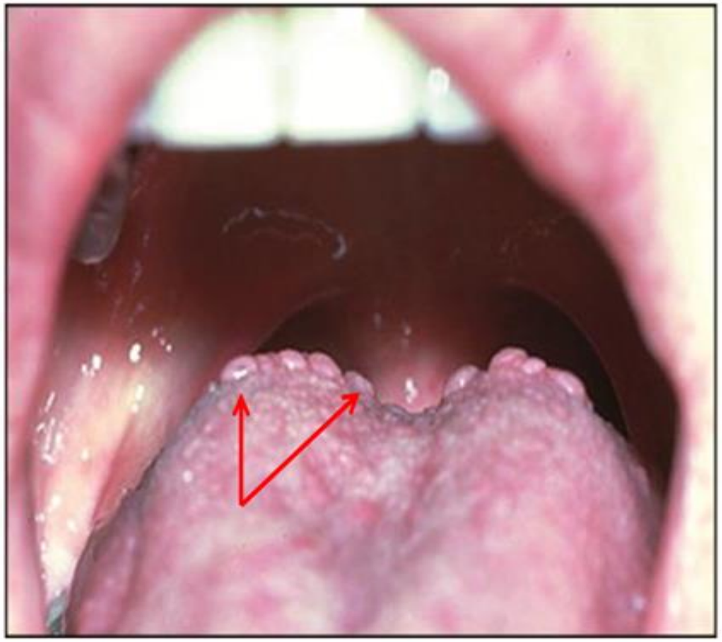
Tongue



oral cavity
taste buds
fungiform papilla
filiform papilla
filiform papilla
filiform papilla
fungiform papilla
lamina propria
intrinsic skeletal muscle fibers

Circumvallate papillae

are largest in size and less numerous papillae - in humans there are between 8 and 12 of them. They are distributed in (V-shaped row in back of tongue), and are surrounded with a groove formed by the in folding of the epithelium. Taste buds are particularly numerous on the lateral surfaces of these papillae. The excretory ducts of serous glands open into the grooves surrounding the papillae (glands of von Ebner). Detect bitter taste

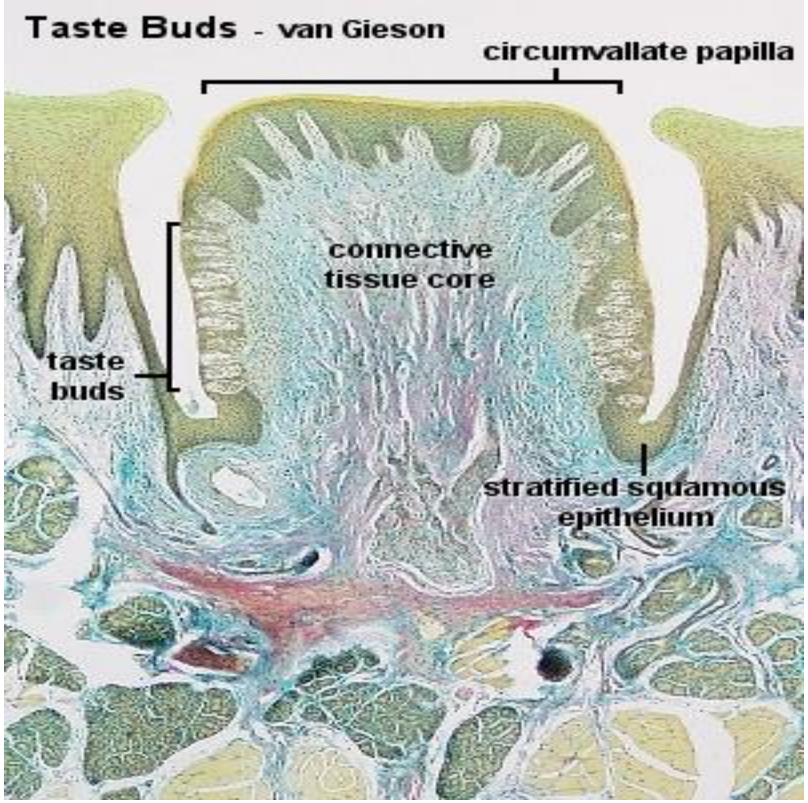


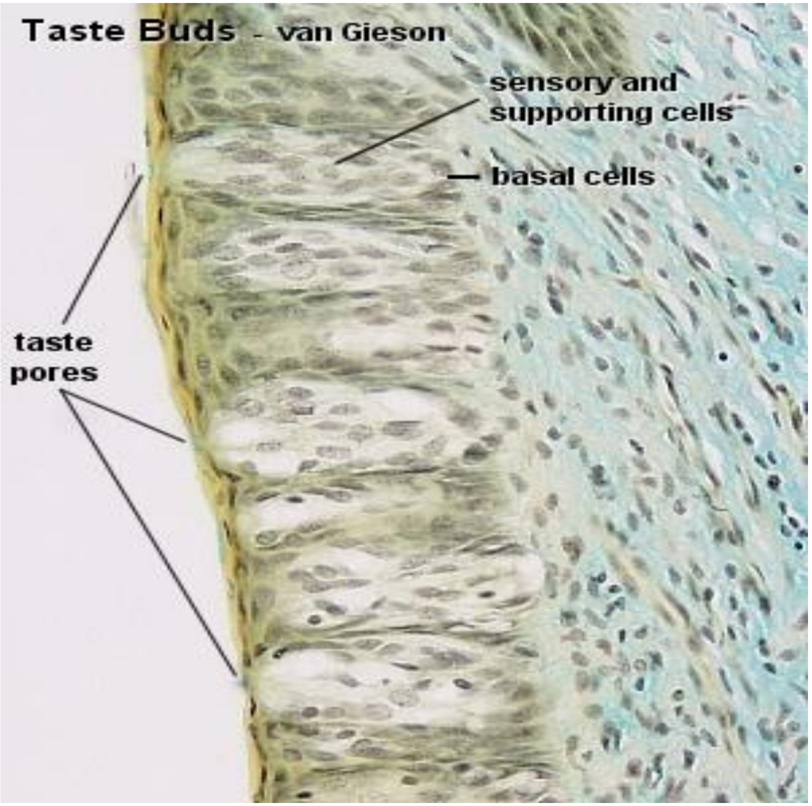
Taste Buds: **Taste is a sensation performed by taste buds receptors located on the tongue**

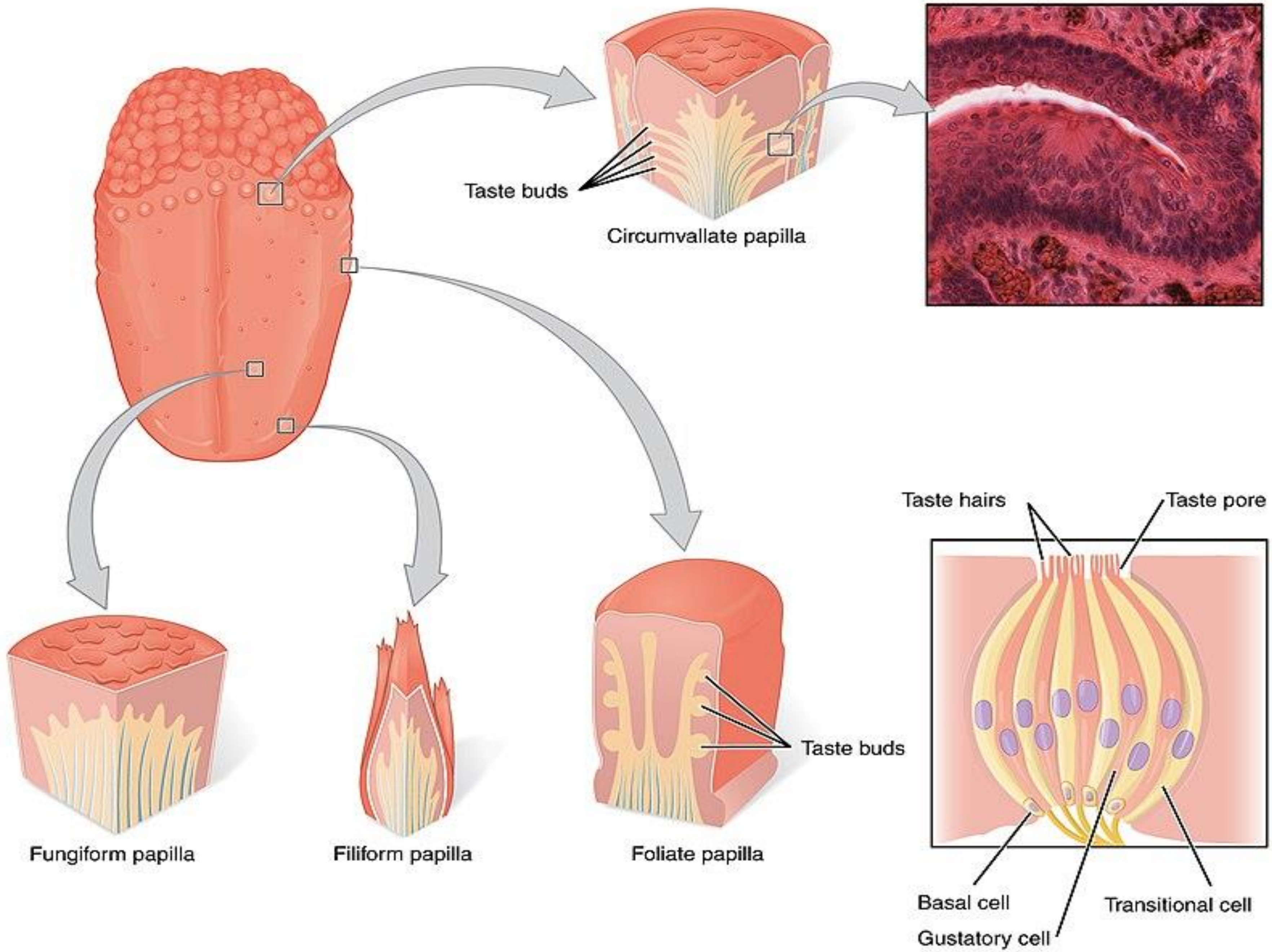
found in circumvallate papillae, fungi form papillae & foliate papillae.

In the smaller numbers on the soft palate, the epiglottis, palatoglossal arches & the posterior wall of the oropharynx.

sweet taste is best appreciated at the tip of the tongue, salt by the area just behind the tip & along the lateral border & bitter taste by circumvallate papillae.





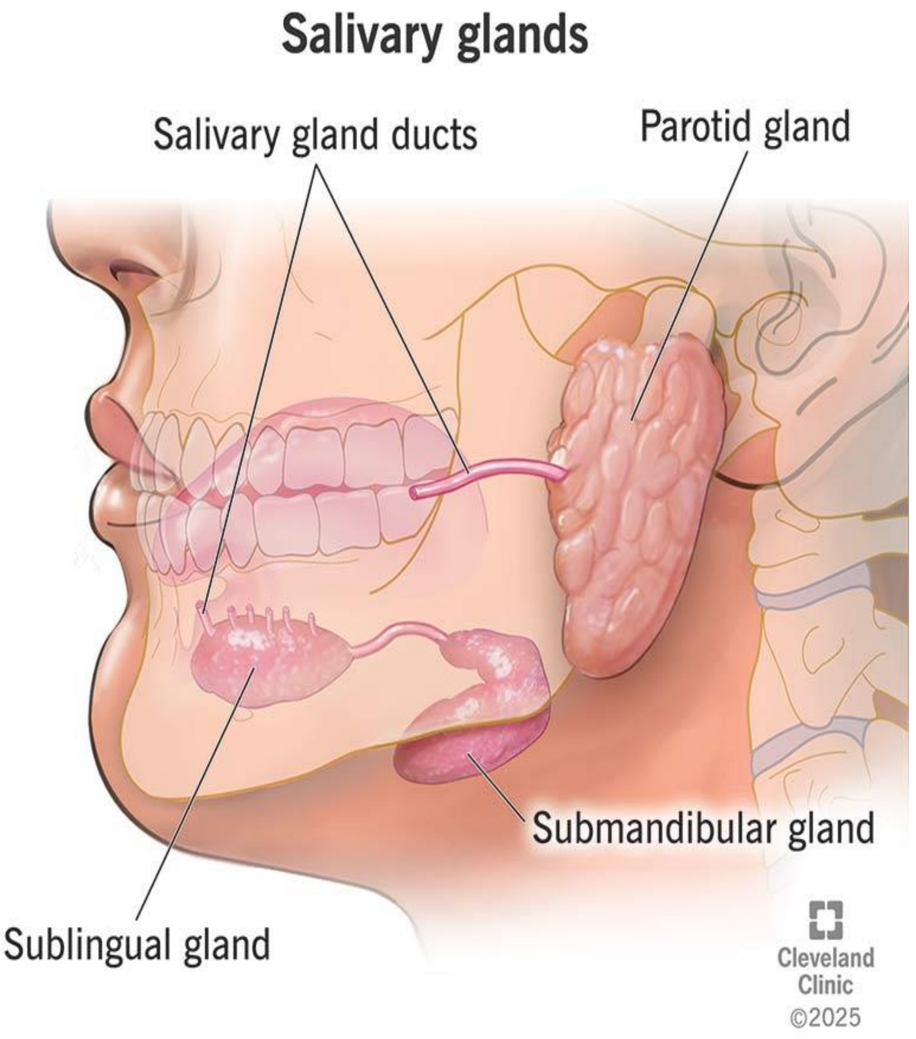

Taste buds
Circumvallate papilla
Taste hairs
Taste pore
Taste buds
Fungiform papilla
Filiform papilla
Foliate papilla
Basal cell
Gustatory cell
Transitional cell

Types of Salivary glands

- **Collectively produce and secrete saliva.**
- **There are 2 groups of s.g.:**
- **The major include:**
- **1-parotid**

  2-submandibular

  3-sublingual
- **The minor : numerous small glands located in the mouth**



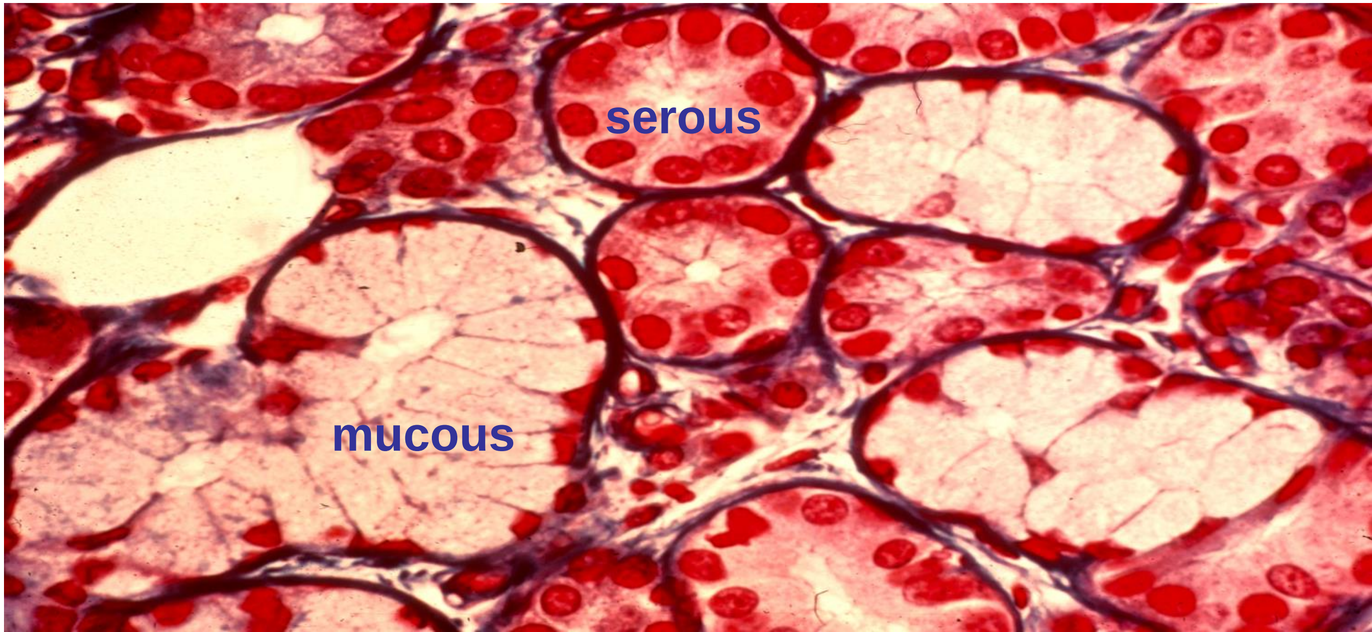
Major salivary glands

- **All of them have capsule-connective tissue send septa divided into lobes— lobules carry blood vessels & nerves like branch of grapes**
- **Consist of :**
- **A-two general types of secretary cells B- duct system**

Cells lining the alveoli

- **A-Secretory cells :1-Serous cells 2-Mucous cells**
- **B-Myoepithelial cells**
- **Serous cells:Are pyramidal in shape, usually form a spherical mass of cells called acinus or alveolus with narrow lumen**
- **Basophilic cytoplasm contain protein secretary granules**
- **they produce protein watery secretion**
- **Mucous cellsAre usually cuboidal to columnar , arranged as tubules**
- **are larger more acidophilic than serous**

secrete mucous



System of ducts

1-Intralobular:small ducts found in lobule they are surrounded by thin layer of c.t.are 2 types:

a-Intercalated ducts

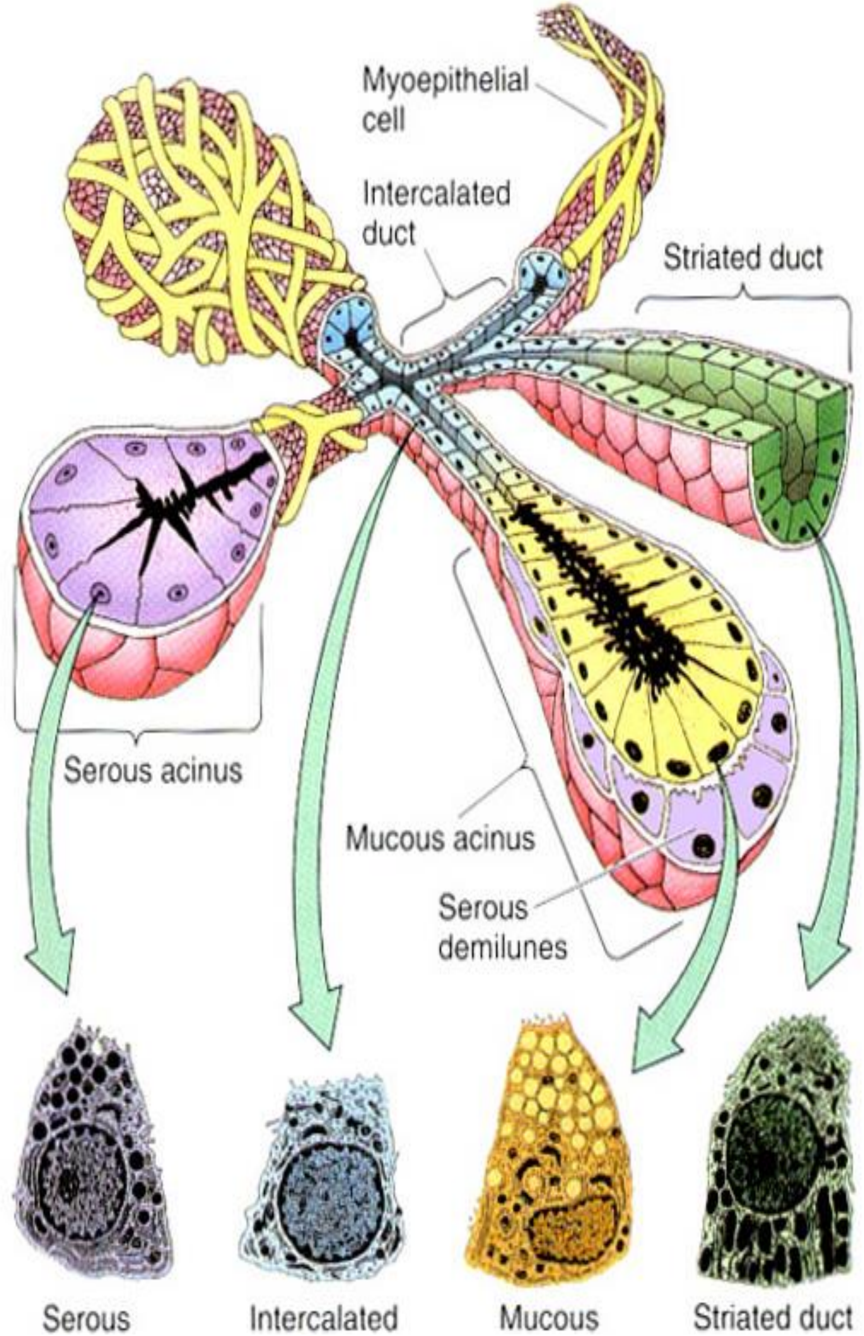
narrow lumen

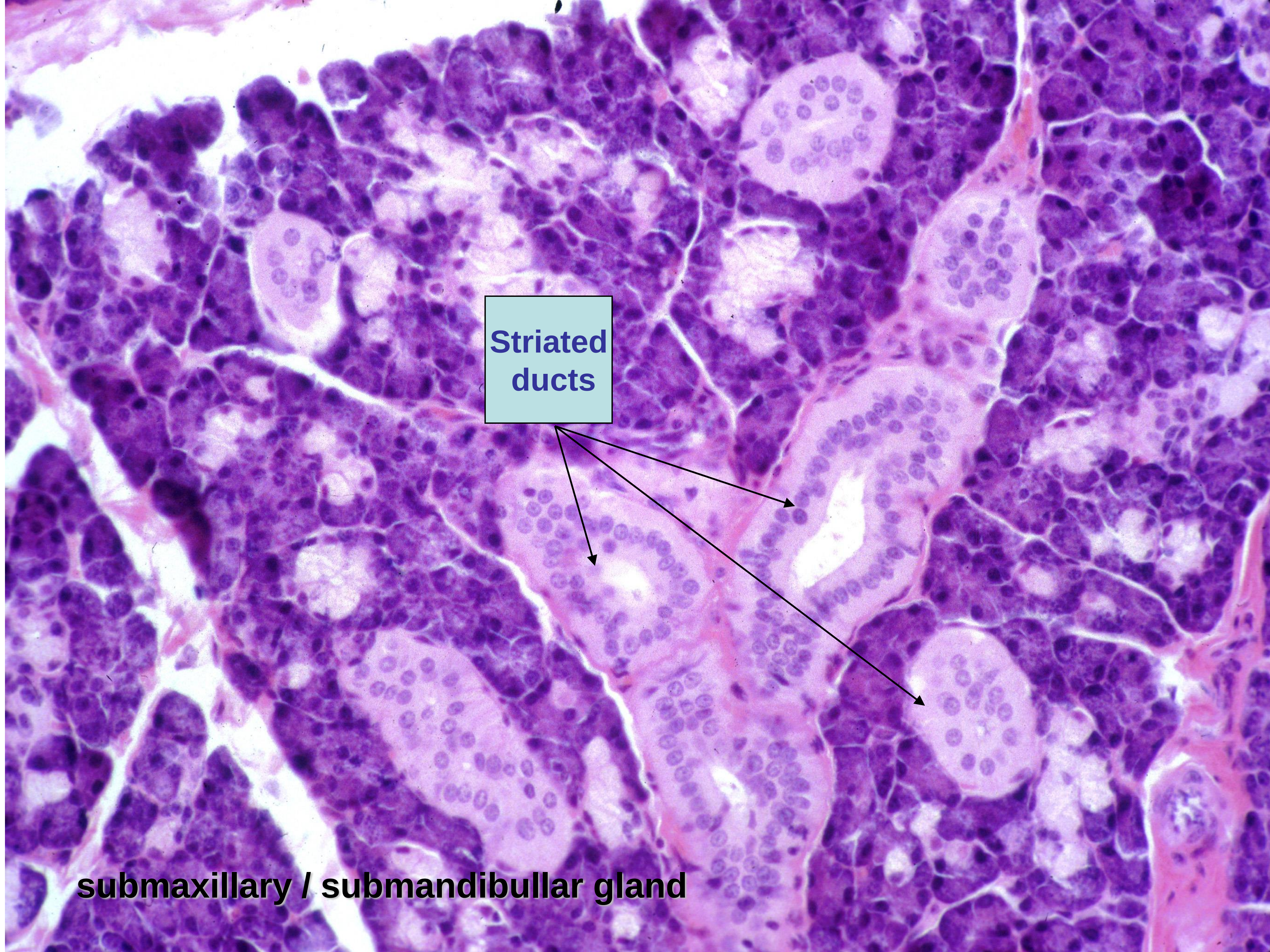
Simple cuboidal ep.

b-Striated ducts:

Has basal striation

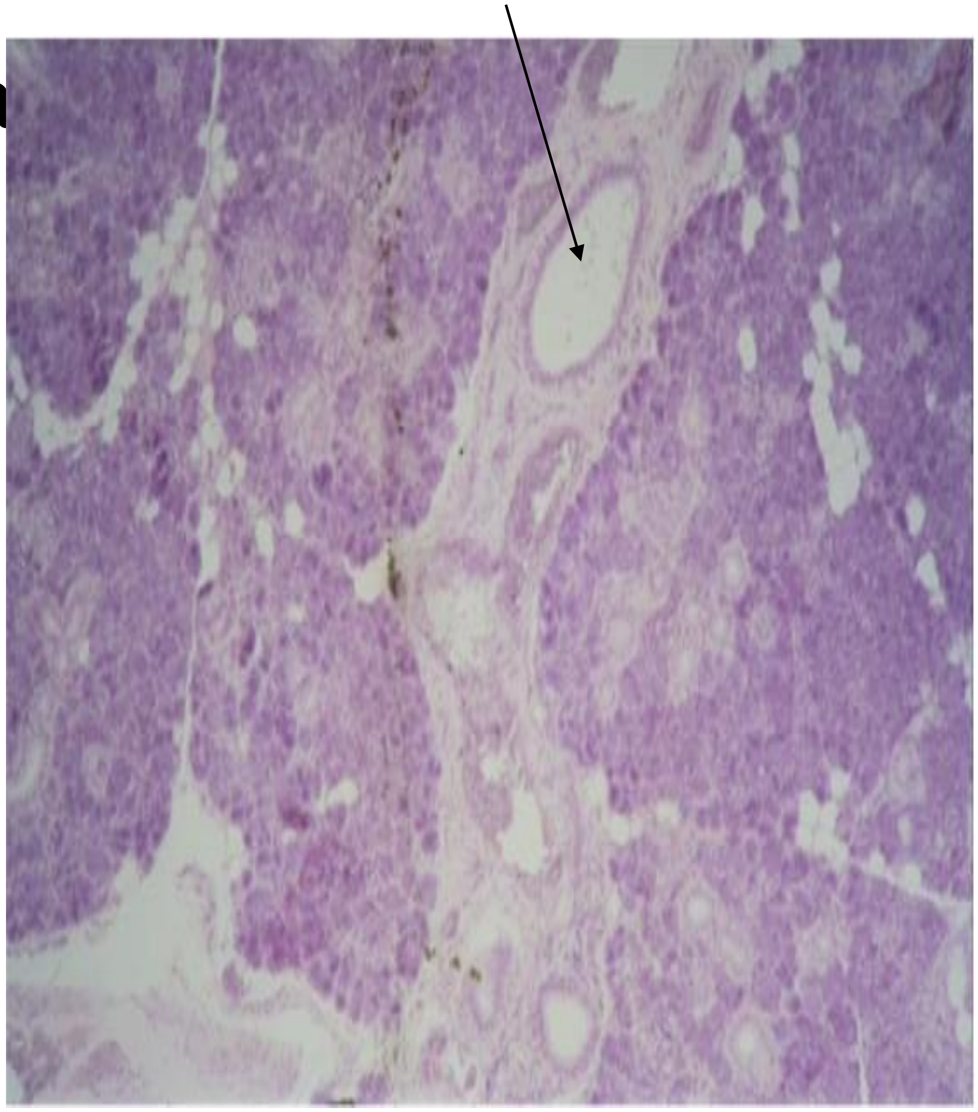
Simple columnar




Striated ducts
submaxillary / submandibullar gland

2-Interlobular ducts

- **Located in the septa between lobules& lobes**
- **Larger than the intra.surrounded by large amount of c.t.**
- **The epithelia vary between simple columnar to stratified**



Parotid gland

- Largest salivary glands.
- Beside the ear lies anterior to the ear
- Their long one duct open into oral cavity opposite the second upper molar tooth on each side
- Are compound acinar gland, are consist of serous
- Produce about 25–30%

  of the saliva

Long intercalated-

&short striated ducts.

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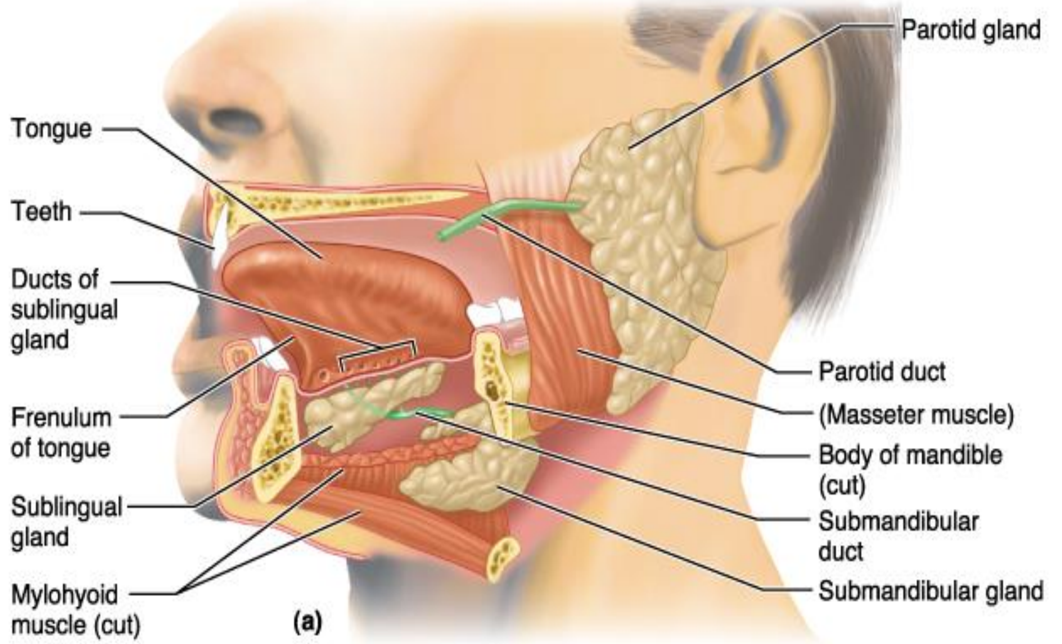
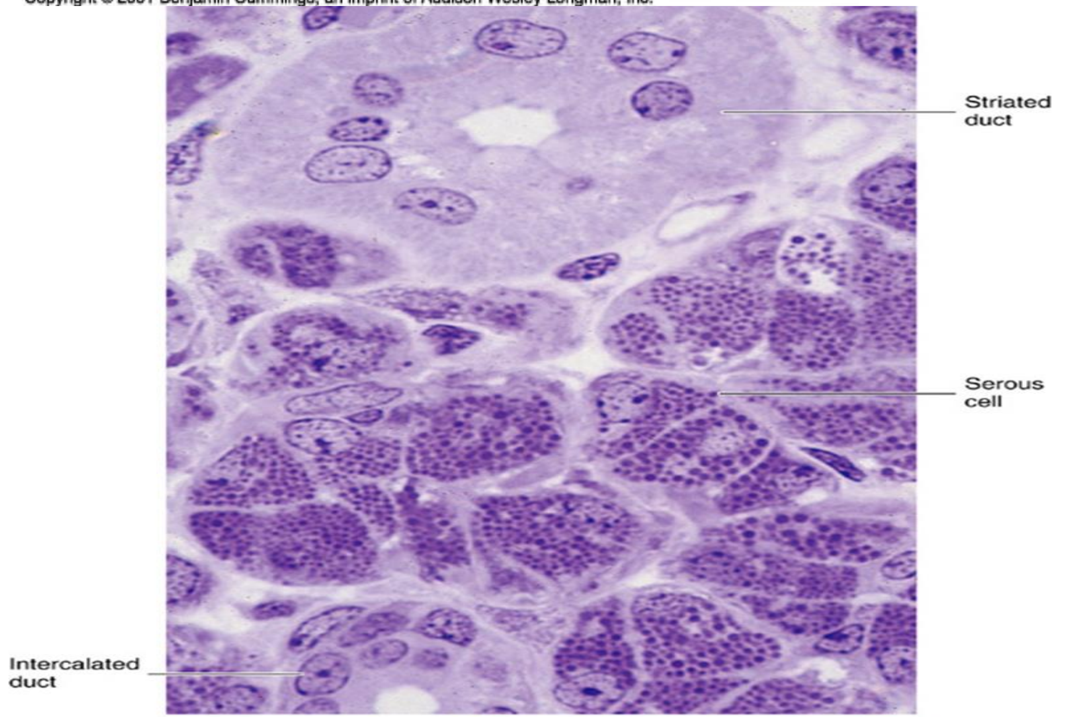
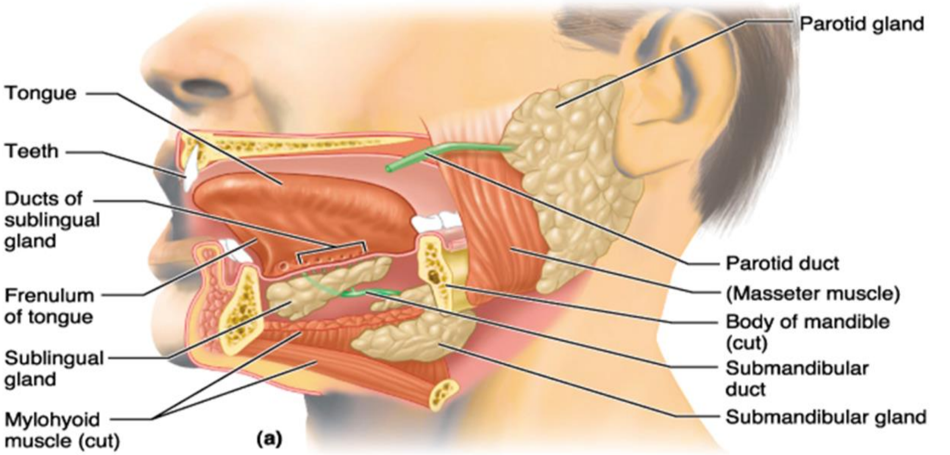
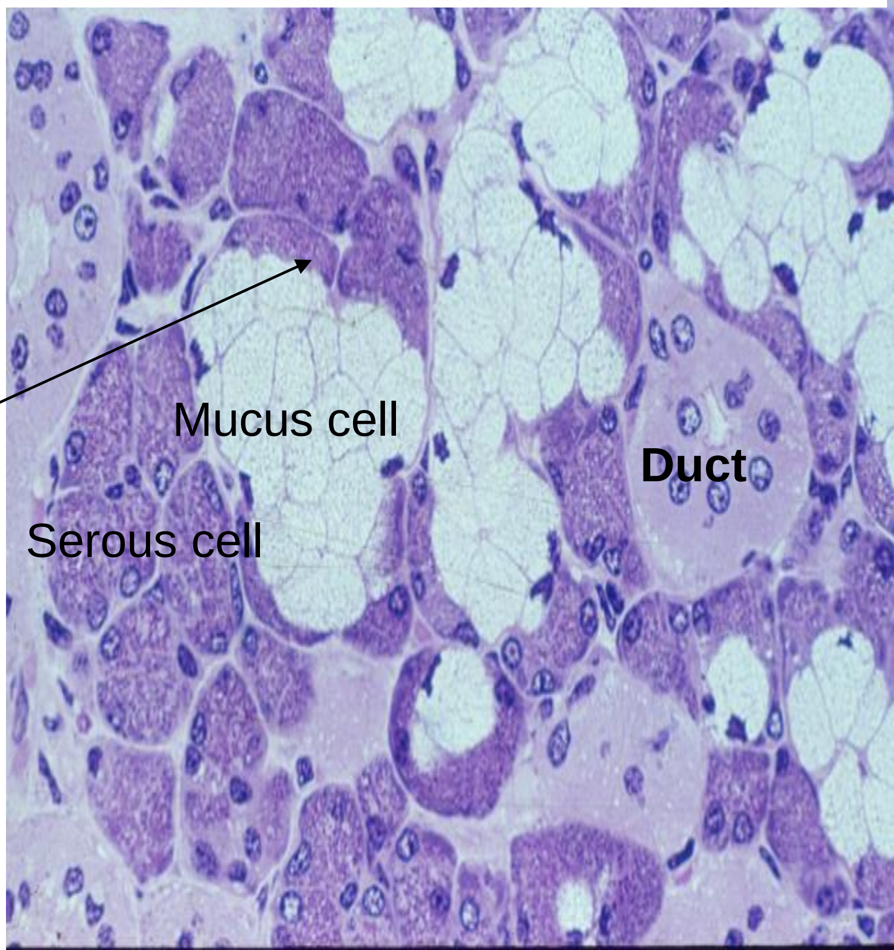




Submandibular gland :

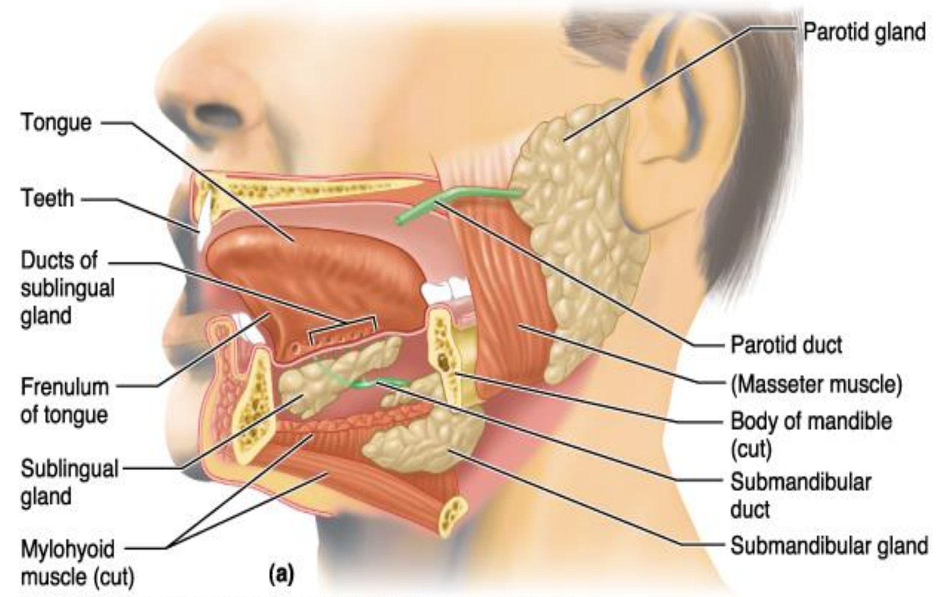
- Compound tubulo-acinar
- Are ovoid shape
- Situated on the either side of neck just below the mandible
- Their duct open into floor of mouth one on each side of the frenulum of tongue
- Secretory portion contain both Serous-mucous, the serous is predominent
- 70%
- Serous demilunes secrete lysozyme
- Long striated while short intercalated ducts

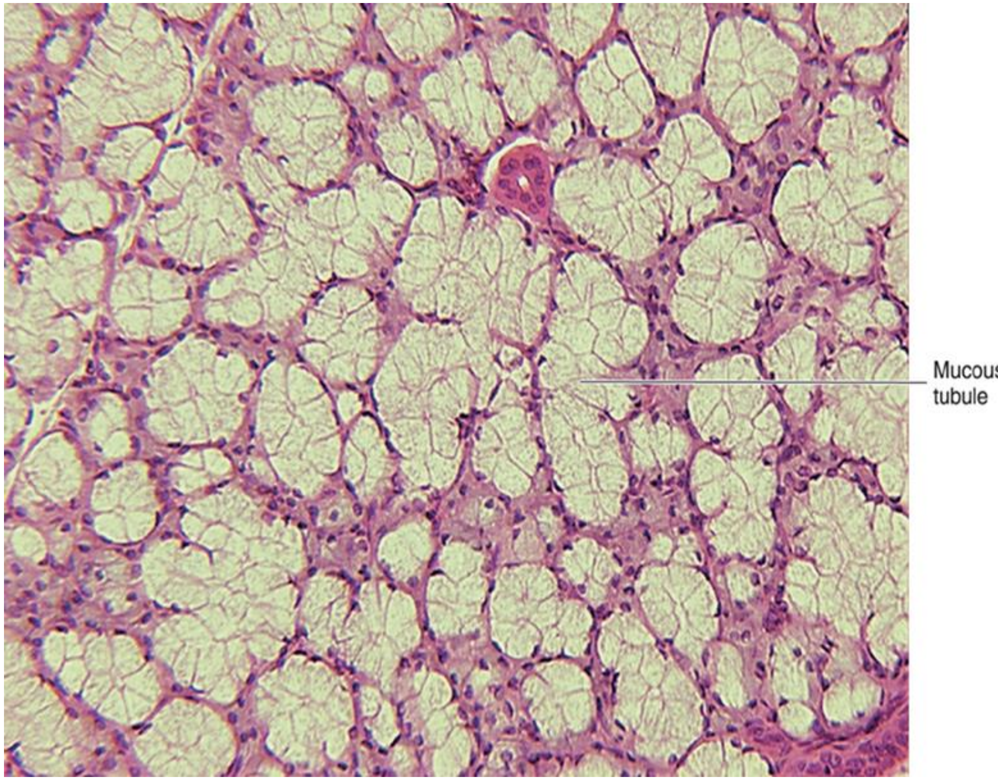




Sublingual gland

- compound tubulo-acinar, are smallest glands
- Located in floor of mouth one on either side of the frenulum of the tongue
- each gland open into oral cavity through 10-12 small ducts
- Contain mucous & serous demilunes
- 5%
- Short intercalated, is ill defined & striated ducts





Thanks